TENSOR PRODUCT WITH A RING SHEAF IS LEFT ADJOINT TO RESTRICTION FOR SHEAVES OF MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tensor-restrict adjunction for presheaves of modules: Hom_{O_1}(G, F_{O_1}) = Hom_{O_2}(O_2 tensor_{p,O_1} G, F). invoked at proof lines 2150-2152. Provides the presheaf-level adjunction the sheaf statement descends from.

Source reference. Original lemma 006R: sources/stacks-sheaves/pinned-original-source.tex, source lines 692–692. Cited at proof lines 2150-2152 as the presheaf adjunction the sheaf statement descends from; load-bearing inside the counted span.

Lemma 0.2. *With X, $\mathcal{O}_1$, $\mathcal{O}_2$, $\mathcal{F}$ and $\mathcal{G}$ as above there exists a canonical bijection*

$$\mathrm{Hom}_{\mathcal{O}_1}(\mathcal{G}, \mathcal{F}_{\mathcal{O}_1}) = \mathrm{Hom}_{\mathcal{O}_2}(\mathcal{O}_2 \otimes_{\mathcal{O}_1} \mathcal{G}, \mathcal{F})$$

In other words, the restriction and change of rings functors are adjoint to each other.

Proof. This follows from Lemma 006R and the fact that $\mathrm{Hom}_{\mathcal{O}_2}(\mathcal{O}_2 \otimes_{\mathcal{O}_1} \mathcal{G}, \mathcal{F}) = \mathrm{Hom}_{\mathcal{O}_2}(\mathcal{O}_2 \otimes_{p,\mathcal{O}_1} \mathcal{G}, \mathcal{F})$ because $\mathcal{F}$ is a sheaf. □

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